Cycle-Jump Demonstration Using a Chaboche Unified Viscoplastic UMAT

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1.1 Motivation

Long cyclic simulations are computationally expensive when every loading cycle is resolved explicitly. A cycle-jump approach reduces the cost by identifying a stabilized per-cycle evolution of selected internal variables and extrapolating this evolution over many cycles. In the present demonstration, the cycle-jump variable is the accumulated viscoplastic strain stored by the UMAT as SDV1.

1.2 Abaqus–UMAT Model

The model was implemented in Abaqus/Standard using a single C3D8 block element smoke-test geometry. The material response was described by a Chaboche-v1 unified viscoplastic UMAT and loaded under displacement-controlled fully reversed cyclic loading. The state variable SDV1 represents the accumulated viscoplastic strain p, while SDV15 stores the last viscoplastic increment. This compact setup was selected to isolate the UMAT response, the Abaqus interface, and the cycle-jump postprocessing workflow.

1.3 Amplitude Selection

An amplitude sweep was first performed to identify a physically useful validation amplitude. The $\pm 0.1\%$ and $\pm 0.2\%$ cases remained elastic or near-elastic, whereas the $\pm 0.5\%$ case activated viscoplasticity at a moderate stress level. The $\pm 1.0\%$ case produced stronger plastic cycling. Therefore, the $\pm 0.5\%$ strain amplitude was selected for the cycle-jump demonstration.

1.4 Explicit 10-Cycle Baseline

The selected $\pm 0.5\%$ case was simulated explicitly for 10 cycles. The analysis completed with 507 increments, 0 cutbacks, 0 warnings, and 0 errors. The simulation produced nonzero stress, nonzero reaction force, and monotonic accumulated state-variable evolution. The selected hysteresis loops show that the loop shape becomes nearly repeatable after the first-cycle transient.

1.5 Stabilized Internal-Variable Increment

The total SDV1 value is cumulative because it represents accumulated viscoplastic strain. It should therefore increase monotonically and should not be used directly as a stabilization criterion. Instead, stabilization was assessed using the per-cycle increment ΔSDV1. Cycles 2–10 were used as the stabilized reference window. Over this window, the mean increment was 0.007185465191, the standard deviation was $3.368213202 \times 10^{-6}$, and the relative range was 0.001429037192, corresponding to approximately 0.1429%.

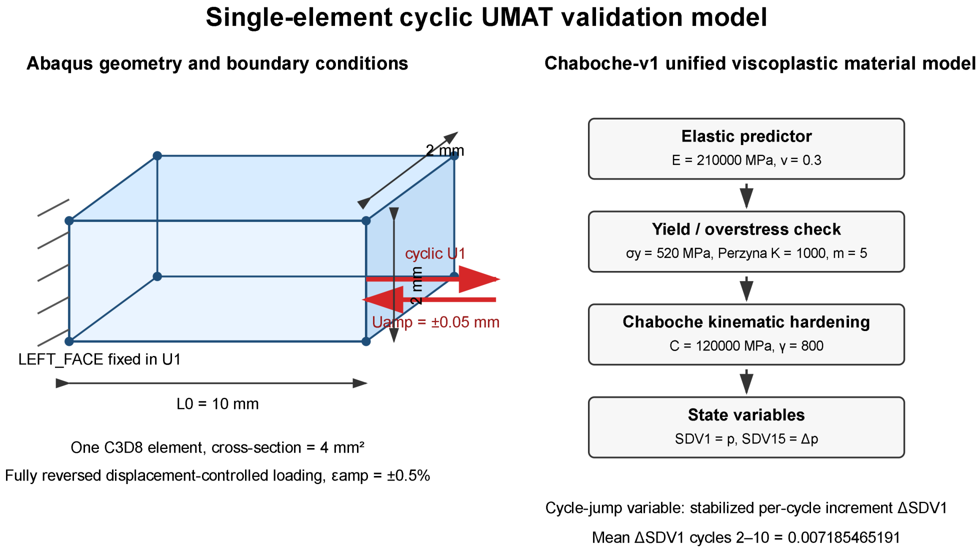

Figure 1: Single-element Abaqus validation model and Chaboche-v1 material-state workflow. The block is a one-element C3D8 specimen with $L_0 = 10$ mm, a 2 mm $\times$ 2 mm cross-section, fixed left face, and displacement-controlled cyclic loading on the right face.

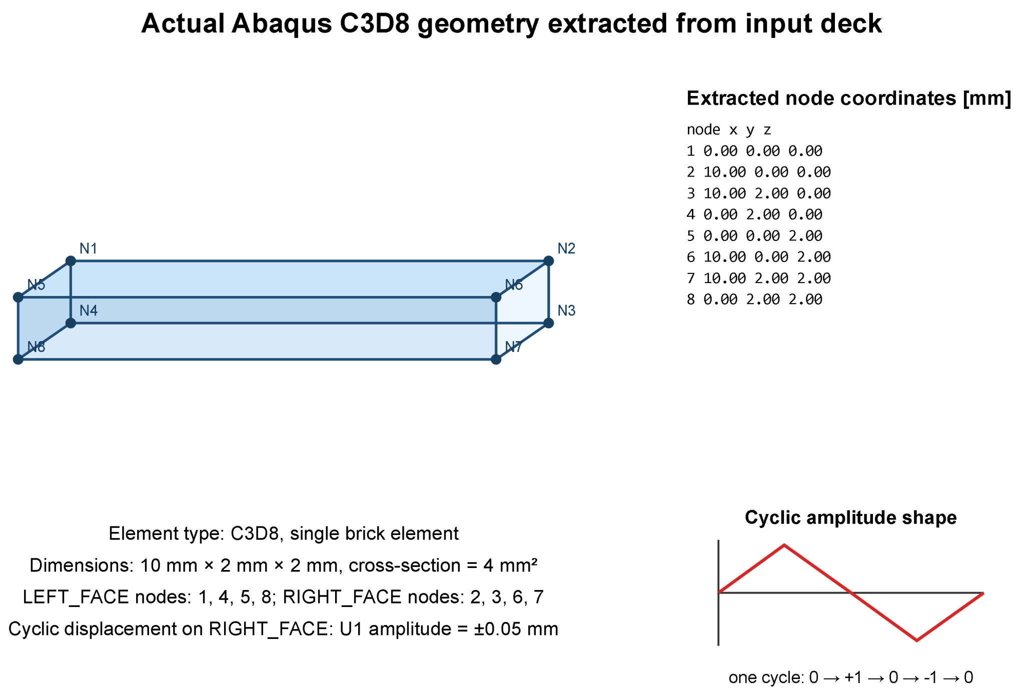

Figure 2: Actual Abaqus single-element geometry reconstructed from the input deck using the node coordinates and C3D8 element connectivity. The extracted model confirms the 10 mm $\times$ 2 mm $\times$ 2 mm block, the left and right node sets, and the displacement-controlled cyclic loading direction.

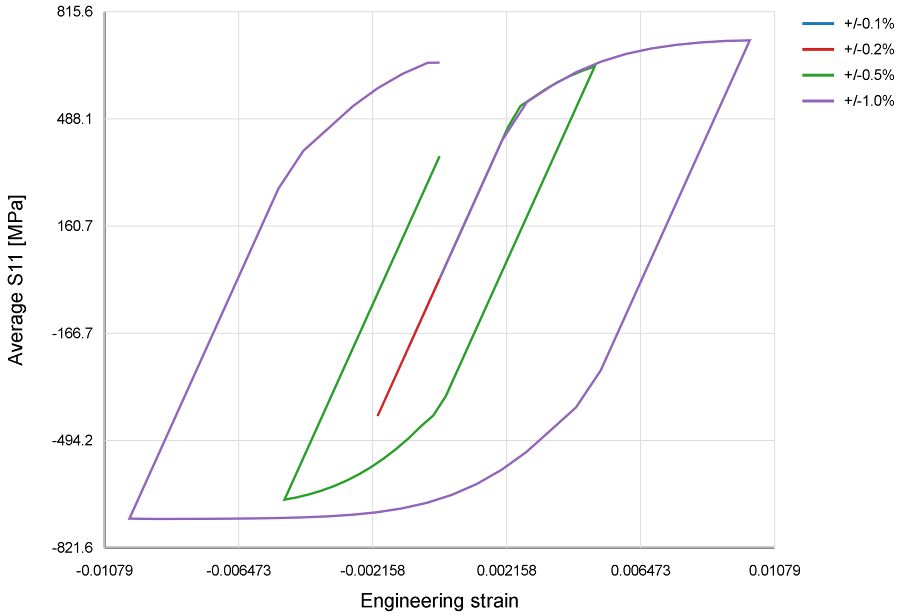

Figure 3: Amplitude sweep for the Chaboche-v1 UMAT. The smaller amplitudes remain mainly elastic, while the ±0.5% case activates viscoplasticity at a moderate stress level and was selected for the cycle-jump demonstration.

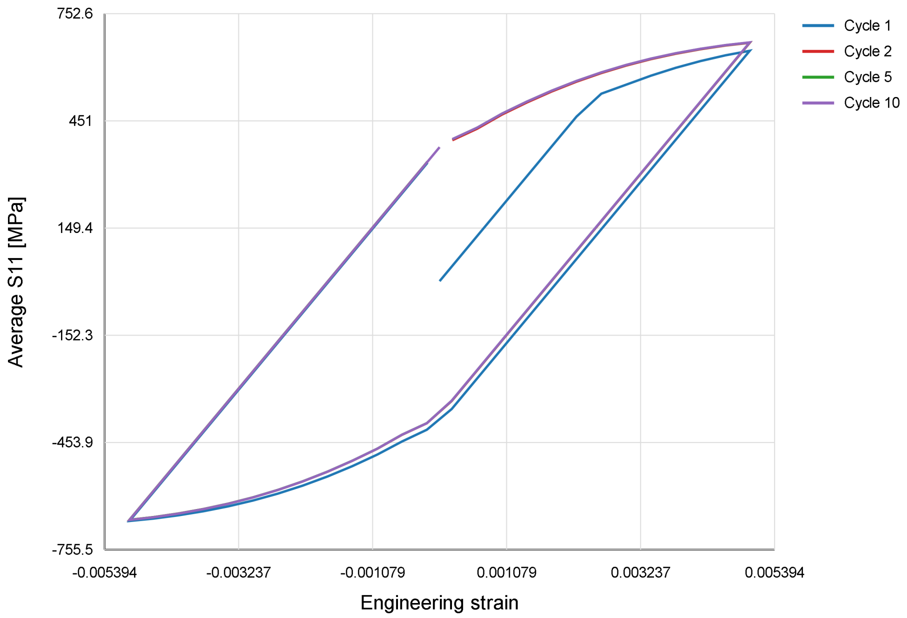

Figure 4: Selected hysteresis loops from the explicit 10-cycle Abaqus simulation at ±0.5% strain amplitude. The loop shape becomes nearly repeatable after the initial transient.

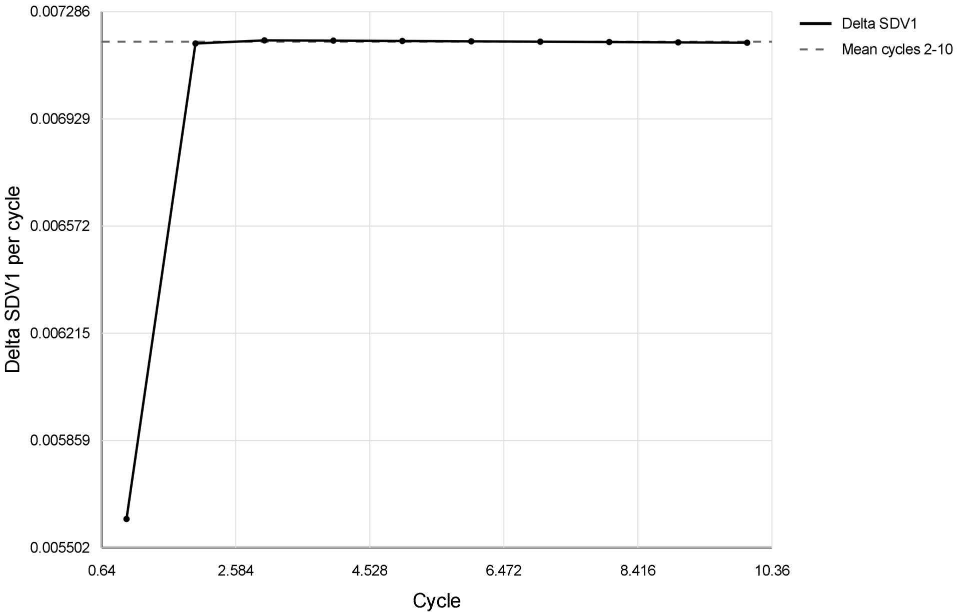

Figure 5: Per-cycle increment of accumulated viscoplastic strain. After the first cycle, the increment remains nearly constant, supporting the use of a cycle-jump predictor.

1.6 Cycle-Jump Predictor

The postprocessing-level cycle-jump predictor was defined as

$$p_N^{\text{pred}} = p_{10} + (N - 10)\,\overline{\Delta p}_{2-10},$$

where p_N^{pred} is the predicted accumulated viscoplastic strain at cycle N, p_{10} is the explicit value at cycle 10, and $\overline{\Delta p}_{2-10}$ is the mean per-cycle increment from cycles 2–10. The resulting predictions were $p_{20} = 0.1421214351$, $p_{50} = 0.3576853909$, $p_{100} = 0.7169586504$, $p_{200} = 1.435505169$, $p_{500} = 3.591144727$, and $p_{1000} = 7.183877322$.

1.7 Explicit 20-Cycle Validation

A separate explicit 20-cycle Abaqus simulation was performed to validate the cycle-jump prediction. This analysis completed with 1007 increments, 0 cutbacks, 0 warnings, and 0 errors. The predicted SDV1 value at cycle 20 was 0.1421214351, while the explicit Abaqus result was 0.1420256943. The absolute error was $-9.574084894 \times 10^{-5}$, corresponding to a relative error of 0.0674%.

1.8 Discussion and Limitations

This result validates a postprocessing-level cycle-jump predictor for the simplified Chaboche-v1 model and the selected ±0.5% strain-amplitude test case. The method does not yet inject jumped STATEV values into Abaqus and is not yet a calibrated fatigue-life model. The next implementation step would be state-variable injection using SDVINI, restart analysis, or another Abaqus workflow that initializes the material state after a cycle jump.

The Chaboche unified viscoplastic UMAT has therefore been successfully implemented and validated at the computational workflow level. It reproduces nonzero stress, reaction force, viscoplastic strain accumulation, cyclic hysteresis, and a stable per-cycle internal-variable increment suitable for cycle-jump prediction. However, the current Chaboche-v1 parameter set

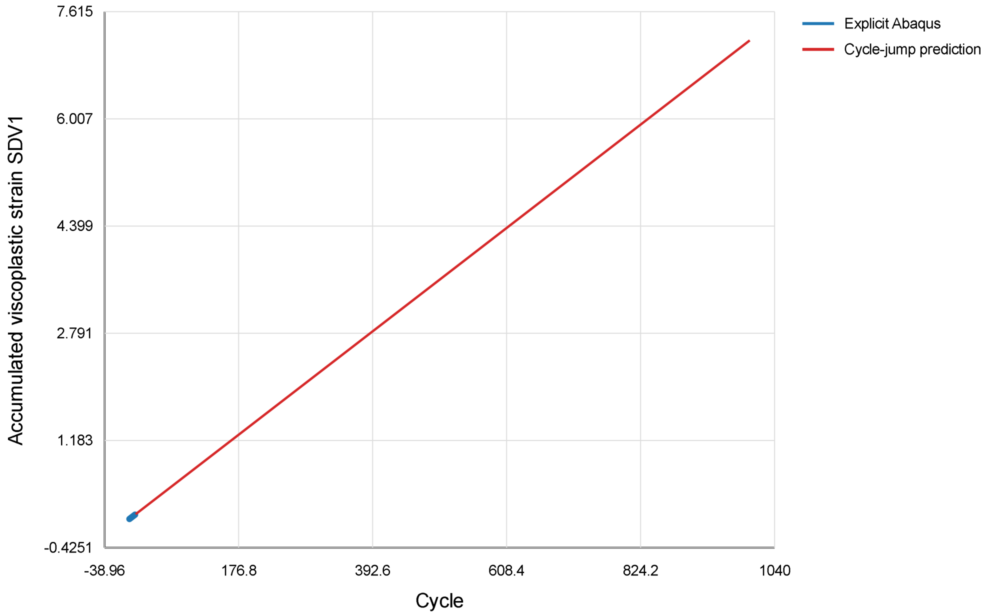

Figure 6: Cycle-jump prediction of accumulated viscoplastic strain using the stabilized per-cycle increment obtained from explicit cycles 2–10.

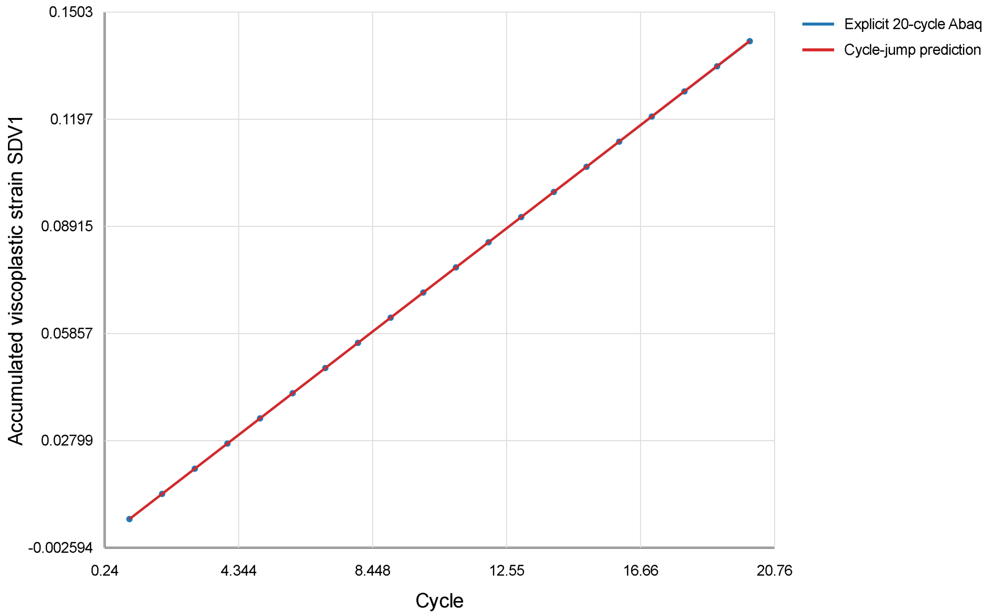

Figure 7: Validation of the cycle-jump predictor against an independent explicit 20-cycle Abaqus simulation. The predicted and explicit SDV1 values agree closely, with a relative error of 0.0674% at cycle 20.

should be treated as a demonstration and validation set, not yet as a fully calibrated 316 stainless steel material model.

1.9 Summary

The Chaboche-v1 UMAT, cyclic Abaqus workflow, postprocessing scripts, and cycle-jump predictor were validated. Explicit cycles 2–10 provided a stabilized per-cycle increment of accumulated viscoplastic strain, and the explicit 20-cycle check confirmed the prediction accuracy with only 0.0674% relative error in SDV1 at cycle 20.

2 Level-2 Preparation: Full-State Cycle-Jump Diagnostics

This section synthesizes the progression from Level-1 scalar SDV1 cycle-jump postprocessing to Level-2 preparation diagnostics for the simplified Chaboche-v1 unified viscoplastic UMAT. It explains the current findings and the rationale for deferring full STATEV injection to a later phase.

2.1 STATEV Inventory

The Chaboche-v1 UMAT manages 15 solution-dependent state variables (STATEV):

Index	Symbol	Physical meaning	Category
1	p	Accumulated viscoplastic strain	Active / required
2–4	X_{11}, X_{22}, X_{33}	Backstress normal components	Active / required
5–7	X_{12}, X_{13}, X_{23}	Backstress shear components	Near-zero
8–10	$E_{vp,11}, E_{vp,22}, E_{vp,33}$	Visc. strain normal	Active / required
11–13	$E_{vp,12}, E_{vp,13}, E_{vp,23}$	Visc. strain shear	Near-zero
14	RISO	Isotropic hardening stress	Recomputable
15	DP	Last visc. multiplier increment	Diagnostic

Table 1: STATEV inventory for Chaboche-v1 UMAT. STATEV(1–13) form the independent material state.

2.2 Full-State Cycle-History Extraction

Extract entire state vector STATEV(1–15) at each cycle end. Perform stability classification over cycles 2–10.

2.2.1 Key Findings

- STATEV(1): **stable extrapolation candidate**
- STATEV(2–4): **needs caution** (high cycle-to-cycle variation)
- STATEV(5–7): **near-zero**
- STATEV(8–10): **needs caution**
- STATEV(11–13): **near-zero**
- STATEV(14–15): **recomputable/diagnostic**

Full-state restart/injection cannot blindly extrapolate all components. The normal stress and strain components exhibit higher cycle-to-cycle variability than the scalar accumulated strain, indicating that phase-point consistency and accurate integration are critical for vector-valued cycle jumping.

2.3 Vector-Valued STATEV Cycle-Jump Control

Extend the scalar cycle-jump approach to a vector of active components: STATEV(1), STATEV(2–4), STATEV(Use first-order and second-order Taylor extrapolation with adaptive curvature checking to estimate a conservative global jump size.

2.3.1 Original Result

- Jump base: cycle 10
- Conservative global ΔN: **2**
- Adaptive target cycle: **12**
- Controlling component: **STATEV(2) /** X_{11}
- First-order SDV1 relative error at target: 0.0118%

The vector-valued analysis is more restrictive than scalar SDV1 jumping because the normal backstress components impose tighter constraints on the global jump.

2.4 Exact-Output Diagnostic Branch

The original extraction used nearest available ODB frames at integer cycle-end times. For phase-sensitive components (especially backstress and viscoplastic strain), this introduces time ambiguity. A separate "exact-output" Abaqus run was created with `TIME MARKS=YES` to force exact integer cycle-end frame output.

2.4.1 Exact-Output Results

- Maximum absolute cycle-end time error: 0 (exact frames) vs. original 0.00974
- Cycle-20 STATEV(1): 0.134750679135 (exact) vs. 0.142025694251 (original)
- Relative difference: **5.12%**

Key observation: The `TIME MARKS=YES` directive forced Abaqus to accept smaller time increments to align with requested output times. This altered the accepted increment sequence compared to the original run, causing the UMAT integration to produce a noticeably different cumulative viscoplastic strain by cycle 20.

2.4.2 Exact Vector Result

- Conservative global ΔN: **1** (vs. original 2)
- Adaptive target cycle: **11** (vs. original 12)
- Controlling component: **STATEV(2) /** X_{11} (unchanged)

2.4.3 Interpretation

The exact-output run successfully demonstrated that phase-point ambiguity can be removed by requesting exact output marks. However, the 5.12% difference in cycle-20 SDV1 indicates that the simplified Chaboche-v1 UMAT is **increment-schedule sensitive**. The exact-output run cannot replace the original validation baseline because it represents a different computational path (forced smaller increments) with a measurably different material response. Therefore, the original 20-cycle result remains the main validation reference, while the exact-output branch is used to demonstrate phase sensitivity and the need for caution in vector-valued STATEV cycle jumping.

2.5 Why STATEV Injection is Deferred

2.5.1 Status

- Level 1 (postprocessing SDV1 prediction): validated with 0.0674% error; thesis-ready as demonstration case
- Level 2a (full-state extraction): completed; inventory and stability classification available
- Level 2b (vector-valued control): completed; shows vector approach is more conservative than scalar
- Level 2c (exact-phase diagnostics): completed; reveals increment-schedule sensitivity

2.5.2 Robustness Requirement

1. **Increment-schedule sensitivity**: The exact-output comparison revealed that the Chaboche-v1 UMAT response is sensitive to the number and size of time increments. An injected state at a given cycle could produce different future predictions depending on the integration scheme used in the next Abaqus run.
2. **Vector-valued complexity**: The vector cycle-jump approach requires consistent treatment of 13 independent components. Without first reducing the UMAT's increment sensitivity, injecting a state vector carries high risk of uncontrolled accuracy loss.
3. **Thesis narrative alignment**: The demonstrated Level-2 preparation diagnostics are themselves strong thesis content. Stopping here allows a clear discussion of why material model integration robustness is necessary before advanced continuation methods.

2.6 Distinction Between Method Levels

Level 1 (Postprocessing): Extract computed STATEV history from existing ODB. Fit polynomial or first-order model to cycle-end values. Extrapolate and predict cycle-end state at skipped cycles. Read-only ODB access; no material model changes. Postprocessing-level accuracy.

Level 2 (Preparation): Extract full state vector. Perform stability and sensitivity analysis. Diagnose increment-schedule dependence. Design injected-state continuation logic. Diagnostic runs only; no material model changes. Diagnostic-level accuracy.

Level 3 (Full Integration): Implement injected-state restart in Abaqus. Modify UMAT or use Abaqus restart keywords. Run accelerated multi-cycle segments with injected STATEV. Active Abaqus workflows and state-variable management. Subject to robustness of UMAT and increment selection.

3 Increment-Schedule Sensitivity of the Chaboche-v1 UMAT

This section freezes the Stage 3 sensitivity study that compared three controlled DMAX settings for the same 20-cycle Chaboche-v1 simulation.

The question is whether accumulated viscoplastic strain remains invariant under time-increment refinement. It does not. The cycle-20 accumulated strain increases monotonically as DMAX decreases, so the UMAT remains increment-size sensitive and Level-3 STATEV injection stays deferred.

Case	DMAX	INC limit	STATEV1 @ cycle 20	Rel. diff. vs ref.	Avg S_{11} @ cycle
Original-output baseline	0.020	4000	0.142025694251	0%	376.434143
DMAX-refined case	0.010	4000	0.143569096923	1.0867%	380.566436
Fine DMAX case	0.005	6000	0.145257070661	2.2752%	385.053924

Table 2: Stage 3 increment-schedule sensitivity summary for the Chaboche-v1 UMAT. The reference value for STATEV1 at cycle 20 is 0.142025694251.

3.1 Cycle-20 Summary

3.2 STATEV1 Versus DMAX

The first sensitivity plot shows the final cycle-20 accumulated viscoplastic strain against the maximum allowed increment size. The trend is monotonic: smaller DMAX produces larger accumulated `STATEV1`.

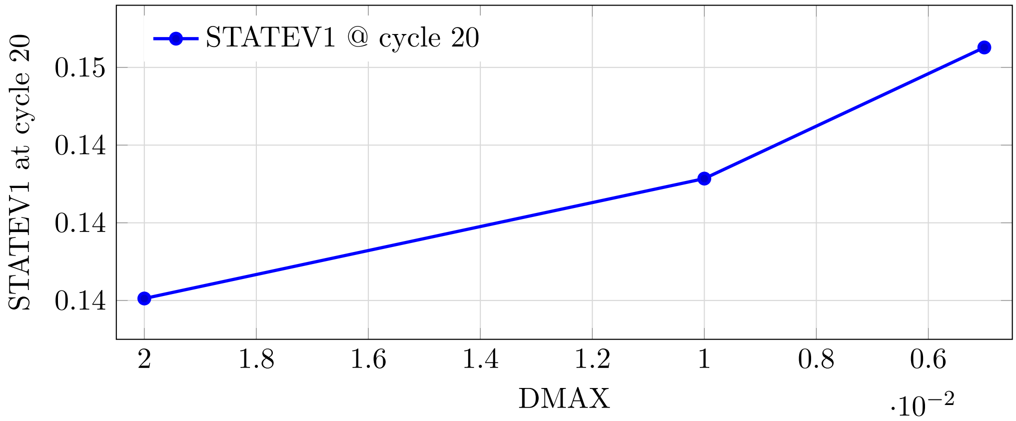

Figure 8: Cycle-20 STATEV1 versus DMAX for the three completed Stage 3 runs.

3.3 STATEV1 Evolution Across Cycles

The second plot shows the cycle-end evolution of STATEV1 for the three completed cases. It confirms that the cases separate progressively as the increment schedule becomes finer. The source SVG versions of the Stage 3 plots are archived in the thesis package figures folder alongside this LaTeX-generated version of the same result.

3.4 Interpretation

The Chaboche-v1 UMAT shows a monotonic increase in accumulated viscoplastic strain as DMAX decreases, confirming increment-size sensitivity. The DMAX $= 0.005$ run required a larger increment limit because the earlier INC $= 2500$ deck failed with too many increments; the corrected INC $= 6000$ run completed successfully and became the final Stage 3 data point. The practical implication is unchanged: Level-3 STATEV injection and a full Nesnas–Saanouni restart-level cycle jump remain deferred until UMAT integration robustness is improved.

4 Predicted-State FE Cycle-Jump Continuation

The preceding sensitivity study identified that the Chaboche-v1 UMAT response depends on the accepted increment schedule. For this reason, the final cycle-jump demonstration was constructed as a controlled Abaqus continuation test rather than as a general production restart

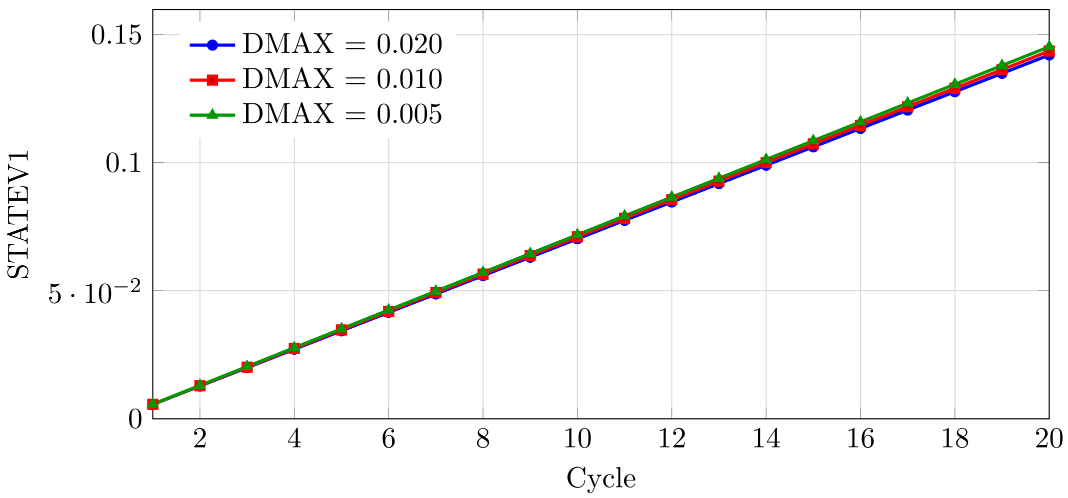

Figure 9: Cycle-end STATEV1 evolution for the three completed increment-schedule cases.

algorithm. The purpose of this section is to document the completed predicted-state FE cycle-jump workflow.

The no-skip Abaqus simulations provide cycle-end histories of the material state and stress. A cycle-space extrapolation is then used to predict a future internal state from the available cycle history. The predicted `STATEV` vector is injected through `SDVINI`, and the predicted residual stress is injected through `SIGINI`. Abaqus then computes only one continuation cycle, and the final state is compared with the corresponding full no-skip Abaqus reference.

4.1 Clean Predicted-State FE Jump

The first predicted FE continuation case used cycle-10 data to predict the cycle-19 state. This predicted state was injected into Abaqus, and one continuation cycle was computed from cycle 19 to cycle 20. The case skipped the intermediate FE cycles 11–18 and reproduced the full 20-cycle reference with sub-percent errors in both accumulated viscoplastic strain and axial stress.

Route	**Skipped**	**Computed route**	**Reference**	**STATEV1 err.**	S_{11} **err.**	**Decision**
Cycle 10 → predicted 19 → 20	8	11 instead of 20	20	0.049427%	0.127013%	Clean success

Table 3: Stage 5B predicted-state FE cycle-jump result. The predicted cycle-19 state was injected with `SDVINI`/`SIGINI` and continued for one computed cycle.

4.2 Target Selection for a Larger Jump

After the clean cycle-19 to cycle-20 result, a 50-cycle no-skip reference was generated and used to scan larger target cycles before running another FE continuation. The scan evaluated the quality of the predicted target state for cycles 29, 39, and 49. The scalar accumulated strain prediction remained accurate for all three targets, but the stress and backstress-related components degraded with increasing jump length. Cycle 29 was therefore selected as the largest acceptable next FE validation target.

4.3 Larger Cycle-29 to Cycle-30 Jump

The second FE continuation case used cycle-10 data to predict the cycle-29 state, then computed only the continuation cycle from cycle 29 to cycle 30. This increased the number of skipped

Target	ΔN	Skipped	Reduction	STATEV1 err. / S_{11} err.	Recommendation
29 → 30	19	18	63.33%	0.135335% / 2.16515%	Acceptable exploratory
39 → 40	29	28	72.5%	0.212701% / 3.30894%	Not headline
49 → 50	39	38	78.0%	0.290737% / 4.45565%	Not headline

Table 4: Stage 6C multi-target prediction scan. The scan used the existing 50-cycle no-skip reference and no additional Abaqus continuation runs.

intermediate FE cycles from 8 to 18. The computed route was reduced from 30 full cycles to 10 base cycles plus one continuation cycle, corresponding to a 63.33% cycle-count reduction.

Route	ΔN	Skipped	Computed route	Reduction	STATEV1 err. / S_{11} err.	Outcome
Cycle 10 → predicted 29 → 30	19	18	11 instead of 30	63.33%	0.0458269% / 2.34366%	Acceptable exploratory success

Table 5: Stage 6D larger predicted-state FE cycle-jump result. The comparison used an interpolated exact-time cycle-30 reference from the 50-cycle Abaqus ODB.

Stage 5B demonstrates a clean predicted FE cycle-jump case. Stage 6D extends the method to a larger jump, increasing skipped intermediate FE cycles from 8 to 18. The accumulated viscoplastic strain remains highly accurate, while the axial stress error increases because stress and backstress prediction becomes the limiting factor for larger jumps.

4.4 Adaptive Jump-Size Selection Inspired by Nesnas–Saanouni

To connect the implementation to the adaptive cycle-jump concept of Nesnas and Saanouni without introducing damage, the original damage variable was replaced by a generic Chaboche control quantity Y_i. For each selected variable, an admissible state change

$$A_i = \tau_i S_i \tag{1}$$

was prescribed and a jump estimate was computed from the ratio of this admissible change to the mean per-cycle increment:

$$\Delta N_i = \left\lfloor \eta \frac{A_i}{|\overline{\Delta Y_i}| + \epsilon} \right\rfloor . \tag{2}$$

Here, η is a safety factor and ϵ prevents division by zero.

The adaptive jump-size controller in this work is inspired by the damage-based cycle-jump strategy of Nesnas–Saanouni. However, the present implementation is not a direct implementation of their coupled damage-viscoplasticity formulation. Since the current Chaboche UMAT does not include an explicit damage variable, the damage-control quantity is replaced by generalized Chaboche state and stress-control variables. Therefore, the contribution of this work is a Chaboche-based adaptive cycle-jump workflow with practical Abaqus continuation through `SDVINI/SIGINI`, rather than a full damage-based fatigue-life model.

Because accumulated viscoplastic strain p is cumulative and was accurately predicted even for larger jumps, it was retained as an accuracy monitor rather than the global limiter. The restart jump size was instead controlled by backstress, viscoplastic strain tensor, and residual stress consistency:

$$\Delta N_{\text{restart}} = \min\left(\Delta N_X, \Delta N_{\varepsilon^{vp}}, \Delta N_S\right). \tag{3}$$

The grouped controller gave $\Delta N_X = 17$, $\Delta N_{\varepsilon^{vp}} = 43$, and $\Delta N_S = 23$, so that $\Delta N_{\text{restart}} = 17$. This recommendation is close to the validated exploratory FE jump with $\Delta N = 19$, for which the Stage 6D continuation skipped 18 intermediate FE cycles and remained acceptable.

The adaptive recommendation was then checked directly in Abaqus by injecting the predicted cycle-27 state and computing the continuation cycle to cycle 28. This Stage 7C validation skipped 16 intermediate FE cycles. The injected state was reproduced accurately in the first output frame, with an absolute error of 1.40×10^{-9} in STATEV1 and 3.39×10^{-6} MPa in S_{11}. At cycle 28, the relative error was 0.0232% for accumulated viscoplastic strain and 2.365% for S_{11}. Therefore, the formula-selected $\Delta N_{\text{restart}} = 17$ was accepted as an exploratory cycle jump, consistent with the Stage 6D result while being selected by the adaptive controller rather than by manual target scanning.

Stage	**Purpose**	ΔN	**Skipped**	**STATEV1 err.**	S_{11} **err.**	**Outcome**
5B	Clean predicted FE jump	9	8	0.049427%	0.127013%	Clean success
6D	Larger manually selected FE jump	19	18	0.0458269%	2.34366%	Accepted exploratory
7B	Grouped adaptive controller	17	16	monitor only	controller only	Recommended
7C	Formula-selected FE validation	17	16	0.0231585%	2.36495%	Accepted exploratory

Table 6: Final cycle-jump validation summary. Stage 7B provides the grouped adaptive recommendation, while Stage 7C validates that recommendation through a direct Abaqus continuation run.

4.5 Interpretation and Limitation

The remaining limitation is not Abaqus initialization, `SDVINI`, `SIGINI`, or UMAT continuation. These mechanisms were verified by the exact-state and predicted-state continuation tests. The remaining limitation is the accuracy of first-order stress and backstress extrapolation for larger cycle jumps.

The core FE cycle-jump concept was successfully demonstrated for the simplified Chaboche-v1 UMAT. A predicted future material state was obtained from cycle-space extrapolation, injected into Abaqus through `SDVINI` and `SIGINI`, and used to continue the FE simulation after skipping intermediate cycles. The method reproduced the no-skip FE reference accurately for accumulated viscoplastic strain and acceptably for stress in the larger exploratory jump.

This result should be interpreted within its intended scope. It does not claim a full Nesnas–Saanouni damage model, a fatigue-life prediction method, a calibrated 316 stainless steel material model, or a general production-ready adaptive cycle-jump framework. Future work should calibrate the Chaboche parameters for the target material, improve the stress and backstress extrapolation beyond the present first-order predictor, and test the grouped adaptive controller on multi-element geometries and longer cyclic histories.